

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI using IBM BOB and IBM watsonx.ai”

Project Tile: AI Travel Planner Agent

Domain of Project: Travel & Tourism

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Problem Statement -

The Challenge

Planning a trip can be a complex and time-consuming process. Travelers often need to search across multiple platforms for destinations, hotels, transportation, tourist attractions, weather information, and budget estimates. The lack of personalized guidance makes it difficult for users to create travel plans that match their interests, budget, travel duration, and preferences. Without an intelligent travel assistant, users may spend more time planning and may not get an optimized travel experience.

The Objective

Build an **AI-powered Travel Planner Agent** that delivers personalized travel planning assistance and recommendations for individuals, families, and business travellers.

The solution should :

Personalized Travel Plans – Recommend customized itineraries based on destination, budget, travel duration, interests, travel type, and number of travellers.

Travel Information Assistance – Provide useful information about destinations, tourist attractions, hotels, transportation options, local food, and travel guidelines.

Trip Planning & Feedback – Allow users to enter their travel requirements and receive instant AI-generated travel plans with recommendations and suggestions.

Smart Travel Guidance – Suggest suitable accommodations, transportation options, packing tips, safety recommendations, and travel advice.

Agentic AI System –

Destination Knowledge Agent (provides destination information and travel insights)

Itinerary Planning Agent (creates personalized day-wise travel schedules)

Budget Optimization Agent (estimates expenses and suggests cost-effective options)

Travel Advisory Agent (provides weather, safety, and travel recommendations)

Travel Dashboard / User Interface – Display generated itineraries, budget breakdowns, recommended places, hotel suggestions, transportation details, and travel tips in an easy-to-understand format.

Proposed Solution:

- The proposed solution is an **AI-powered Travel Planner Agent** built using **IBM BOB**, **IBM watsonx.ai**, and **Meta Llama 3.3 70B Instruct** to provide personalized travel recommendations and itinerary planning.
- The system understands user preferences such as destination, budget, duration, and interests to generate customized travel plans.
- The solution includes four key agents:
 - **Destination Knowledge Agent** – Provides destination details and travel insights.
 - **Itinerary Planning Agent** – Generates personalized day-wise travel schedules.
 - **Budget Optimization Agent** – Estimates expenses and suggests cost-effective options.
 - **Travel Advisory Agent** – Provides safety tips, weather guidance, and travel recommendations.
- Built with a **Flask-based interface**, the system uses IBM watsonx.ai and Meta Llama to deliver intelligent, personalized, and efficient travel assistance.

Technology Used

IBM BOB – For building and managing the Agentic AI workflow.

IBM watsonx.ai – For foundation model execution, AI development, and model governance.

Meta Llama 3.3 70B Instruct – Used for natural language understanding, reasoning, and generating personalized travel recommendations.

IBM Cloud – Provides secure and scalable infrastructure for deploying the AI Travel Planner Agent.

Flask – Used to develop the web application interface for user interaction.

Python – Used for backend development and AI integration.

GitHub – Used for project code management and collaboration.

AI Agent Components Used

- **IBM BOB Agent**
 - Handles user requests and manages the AI workflow.
 - Coordinates travel planning tasks and generates intelligent responses.
- **IBM watsonx.ai Foundation Model**
 - Connects with Meta Llama 3.3 70B Instruct for natural language understanding and travel recommendations.
- **User Input Component**
 - Collects travel details such as destination, budget, duration, and interests.
- **AI Processing Component**
 - Analyzes user requirements and generates personalized travel plans.
- **Response Output Component**
 - Displays itinerary, budget suggestions, attractions, hotels, and travel guidance.
- **Flask Web Interface**
 - Provides an interactive platform for users to communicate with the Travel Planner Agent.

AI Travel Planner Agent Workflow

User Input
(Destination, Budget, Days, Interests)



IBM BOB Agent
(Understands user requirements)



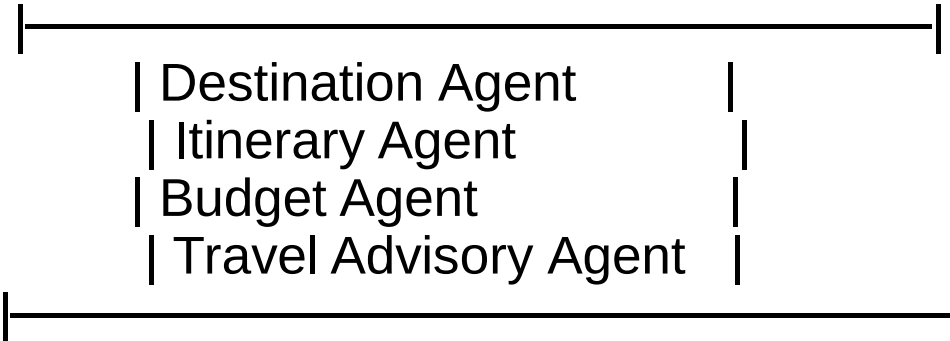
IBM watsonx.ai
(Foundation Model Processing)



Meta Llama 3.3 70B Instruct
(Reasoning & Recommendation)

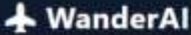


Travel Planning Agents



AI Generated Output
(Itinerary, Budget, Hotels,
Attractions, Travel Tips)

AI Travel Planner User Interface

 Powered by Meta Llama 3.3 & watsonxai

Plan Your Dream Trip

AI-powered personalised itineraries generated using Meta Llama 3.3 70B Instruct in seconds.

Trip Details

 Travelling From e.g. New York, USA	 Destination e.g. Paris, France	
 Travel Dates e.g. Dec 20 – Jan 2	 Nights / Days 5	 Travellers 2
 Total Budget (USD) 2000	 Travel Type Solo	

Approximate total budget for the entire trip.

 **Travel Interests** (select all that apply)

AI Generated Personalized Itinerary



New Trip



Hyderabad → Goa

5 Days

2 Solo

\$ 2000

20 Jul - 24 Jul

Print

Copy

Your Personalised Itinerary

Generated by IBM Granite

Use proper grammar and spelling.

Trip Overview

This 5-day adventure from Hyderabad to Goa promises an unforgettable blend of adventure, outdoor activities, relaxation on pristine beaches, and indulgence in local culinary delights. The trip highlights include exploring the vibrant beaches of North Goa, experiencing water sports like surfing and parasailing, visiting historical sites such as Fort Aguada and the Basilica of Bom Jesus, and enjoying the rich Goan cuisine, known for its seafood delicacies.

Day-wise Itinerary

Day 1: Arrival and Acclimatization

- Morning:** Arrive at Dabolim Airport in Goa. Check into your hotel and spend some time relaxing.
- Afternoon:** Visit Calangute Beach, one of the most popular beaches in North Goa. Enjoy swimming and sunbathing.
- Evening:** Explore the local market near Calangute Beach for shopping and trying street food.

Day 2: Beach Hopping

Trip Summary

From: Hyderabad

To: Goa

Duration: 5 days

Group: 2 (Solo)

Budget: \$2000

Interests: Adventure & Outdoor Activities, Beach & Relaxation, Food & Culinary

Ask WanderAI

Refine your itinerary or ask follow-up questions



Your itinerary is ready! Ask me anything — "Add a day trip to Versailles", "Find cheaper hotel options", or "What should I pack?"

Role of Agentic AI in the Travel Planner Agent

- Agentic AI enables the **Travel Planner Agent** to function as an intelligent, autonomous travel assistant rather than a simple chatbot. Using **IBM BOB**, **IBM watsonx.ai**, and the **Meta Llama 3.3 70B Instruct** foundation model, it understands user travel preferences and generates personalized travel recommendations.
- The AI analyzes inputs such as **destination, budget, travel duration, interests, and number of travellers** to create customized itineraries, recommend attractions, estimate budgets, and provide travel guidance. It can also respond to follow-up questions, making trip planning more interactive and efficient.
- Overall, Agentic AI makes the system **proactive, personalized, and goal-driven**, acting as a smart digital travel assistant that simplifies trip planning and enhances the overall travel experience.

Novelty and Uniqueness

- The proposed **AI Travel Planner Agent** combines **Agentic AI**, **IBM BOB**, and **IBM watsonx.ai** to deliver an intelligent and personalized travel planning experience.
- **Agentic AI-Based Planning** – Uses specialized AI agents to handle destination knowledge, itinerary generation, budget estimation, and travel guidance.
- **Personalized Recommendations** – Generates customized travel plans based on destination, budget, travel duration, interests, and travel type.
- **Interactive AI Assistance** – Supports follow-up questions, enabling users to refine and update their travel plans.
- **Comprehensive Travel Guidance** – Provides day-wise itineraries, hotel suggestions, transportation options, attractions, packing tips, and safety recommendations in one platform.
- **User-Friendly Web Application** – Built with Flask, offering a simple and interactive interface for seamless travel planning.
- **Scalable AI Solution** – Powered by **IBM watsonx.ai** and **Meta Llama 3.3 70B Instruct**, ensuring reliable and high-quality AI-generated responses.
- This makes the solution a smart, personalized, and AI-driven travel assistant that simplifies trip planning and enhances the overall travel experience.

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Git Hub Link

Github URL - <https://github.com/Udaya2k4/ai-travel-planner>

Future Scope

- **1. Real-Time Travel Integration**
- The Travel Planner Agent can be integrated with **Google Maps, weather services, flight booking, and hotel booking APIs** to provide real-time information on routes, weather forecasts, flight schedules, hotel availability, and travel costs, making trip planning more accurate and convenient.
- **2. AI-Powered Voice Assistant & Multilingual Support**
- The system can be enhanced with a **voice-enabled assistant** and **multilingual support**, allowing users to plan trips using voice commands and interact in their preferred language. This will improve accessibility and provide a more personalized travel planning experience for users worldwide.

Certificates Earned:

In recognition of the commitment to achieve
professional excellence



Udaya Pasula

Has successfully satisfied the requirements for:

Getting Started with Artificial Intelligence



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Completion date: 08 Jul 2026 (GMT)

Learning hours: 30 mins

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